

Experiment Proposal

Use of aural cues to support spatial mapping in virtual environment

As mentioned in the previous assignment, the spatial properties of an interior space influence human physiological and psychological response. For this reason, if people are trying to attain a holistic understanding of an interior space through a virtual interior tour, as many are today, we would like to provide ways to minimize the human spatial percept error that occurs in VR (due to portraying simulated three-dimensional space in two dimensions). Because there has been research implying that use of hearing and other senses apart from the visual have been effective in providing compensatory cues that improve object memory and sense of presence in the virtual environment context where visual information is not as rich, the motivation for the current experiment is to use non-visual senses to improve an individual's spatial understanding of the interior in a virtual tour. We have chosen to use audio specifically, since compared to the other senses (taste, touch, smell) it is currently the easiest to incorporate as part of a VR environment.

To integrate the virtual tour factor into an experiment that examines whether the use of spatially-informative audio can improve an individual's spatial understanding of a virtual environment, we would like to propose the following controlled experiment. The proposed experiment utilizes a 1x2 design where the two conditions (independent variables) are binaural, spatialized audio and non-binaural, unspatialized audio. Binaural audio in the physical world occurs when localized sound in an environment is recorded using two microphones, creating a 3D stereo sound sensation for a listener using headphones. The definition of "audio spatialization" is operationalized in the context of this study as regulating the gains of left and right ear contributions of an audio source in the space such that they account for distance, angle, physics-based attenuation, and air absorption. However, because it is a virtual tour the binaurality and spatialization of the audio will be simulated through the use of HRTF (head-related transfer function) and physics-based digital signal processing algorithms.

The general idea is to have participants first experience a virtual tour of an interior space with discrete furniture objects via a VR HMD (head-mounted display), and then measure the spatial understanding of the participants through an object mapping task. Measurement will be done in two forms (dependent variables): object positioning error and task completion time. The conditions are manipulated through the spatialization or non-spatialization of the audio sources that are attached to every piece of furniture in the virtual tour, and all sources must be played for the participant to be considered to have completed the tour. This is to assure that every participant has received the same amount of spatial cues (or not) before progressing to the mapping task, which is an evaluation of their spatial understanding.

We propose a within-subjects design with 20 participants for this experiment. Originally a between-subjects design was considered, due to concern about cognitive fatigue induced by repeated spatial visualization mapping. However, after piloting the study we found that differences between individuals regarding spatial ability as well as their impression of and comfort in a virtual environment tended to be great enough that analysis would likely benefit more from a design that accounts more changes within each individual, therefore transitioning to a within-subjects design.

We hypothesize that due to our motivation and implication from past research, the results of such an experiment would demonstrate a difference between the specified conditions in the measured performance of participants' spatial understanding - in that subjects of the spatialized audio condition will have better performance than the unspatialized condition, shown by shorter task completion time and smaller distance error (object positioning accuracy).

The procedure of the experiment involves having the participant first complete a survey about their experience regarding commonly used technology products as well as VR - this may be helpful in later analysis if we notice differences in the effects of audio cues on spatial understanding of people who are more experienced in VR technology compared to those who are not. Afterwards, the participant will proceed to tour the virtual room by using the HMD; there are multiple pieces of furniture as well as 5 identical chairs placed throughout the room. Each furniture has an audio source containing descriptive speech introducing its role in the room, and the participant selects the piece of furniture using a laser pointer from the VR controller to play the audio source. After a furniture's audio source is played once the furniture turns into a uniform color, and once all pieces of furniture have turned into the same color the participant is considered to have completed the virtual tour. After the virtual tour is completed the participant is presented with a to-scale map of the room they have just explored, including all pieces of furniture correctly positioned and to-scale except for the 5 chairs, which are excluded. The participant is then asked to draw the positions of the 5 chairs in the room and is timed to completion. The same tasks starting from touring the virtual room are then repeated, but with a different room that has different arrangement of furniture and coloring characteristics. This concludes the actual procedure. We then measure the distances from the participant-mapped positions of the 5 chairs to the their correct positions to get multiple measurements of participant error in object mapping.

Some potential ways to visualize the data are included below. Figure 1 is a way to visually determine a difference (if existing) between the two conditions in the study. In a real example this would of course include error bars to determine if the difference is actually significant. The mean accuracy is intentionally calculated from the multiple measurements to get a more representative value than if the participant only had the chance to map one object in the room, which could have been particularly easy or difficult to map due to its location. Figure 2, instead of visualizing the mean accuracy, visualizes the individual cm distance error for each of the object measurements taken (out of 5 total). This can be useful to determine if there exists any chair that has significantly different spatial error when compared to the rest, and then we can examine its specific properties that may have caused it to be the outlier, for example being located in what is apparently a more difficult-to-spatialize part of the room.

Figure 1.

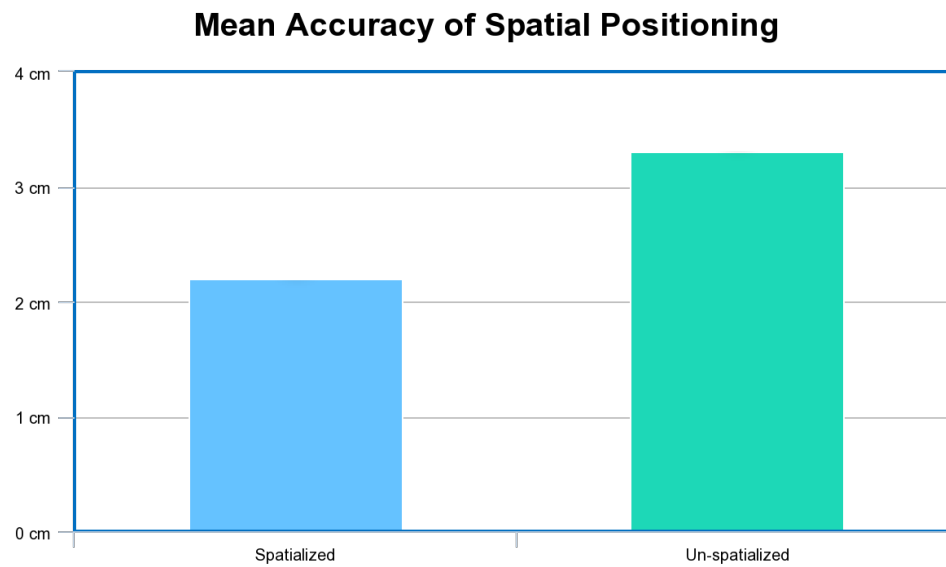


Figure 2.

